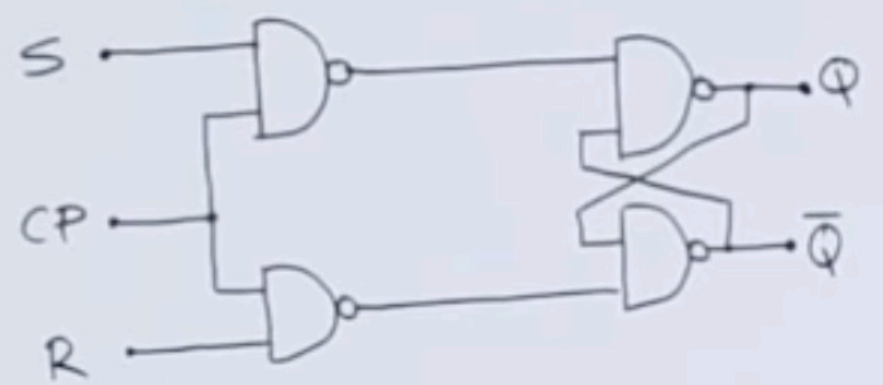


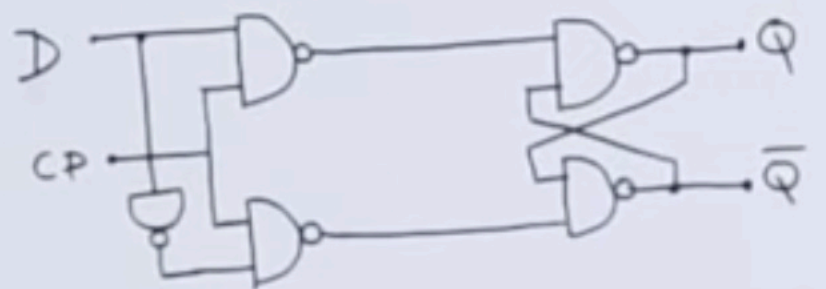
Summary of Flip-Flops

Logic diagram

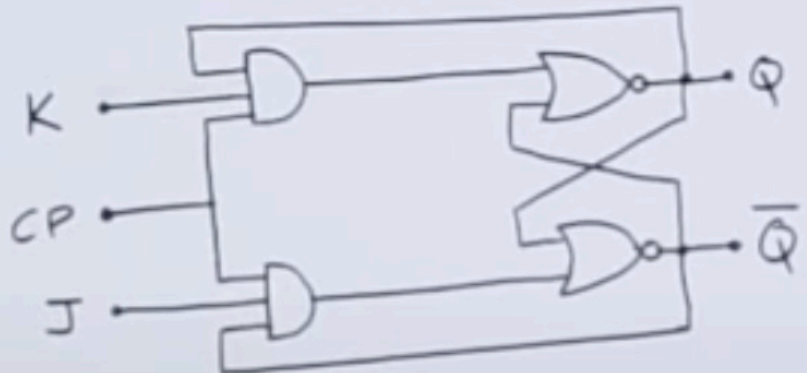
S-R F/F



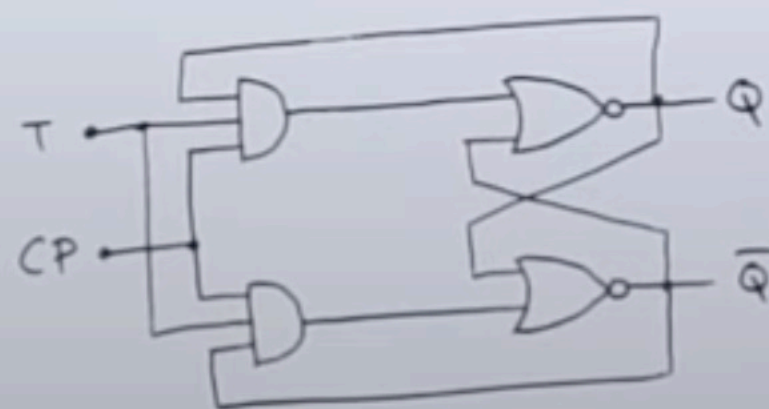
D F/F



J-K F/F



T F/F



Characteristic Table

S	R	Q(t+1)
0	0	Q(t) No Change
0	1	0 Reset
1	0	1 Set
1	1	? Indeterminate

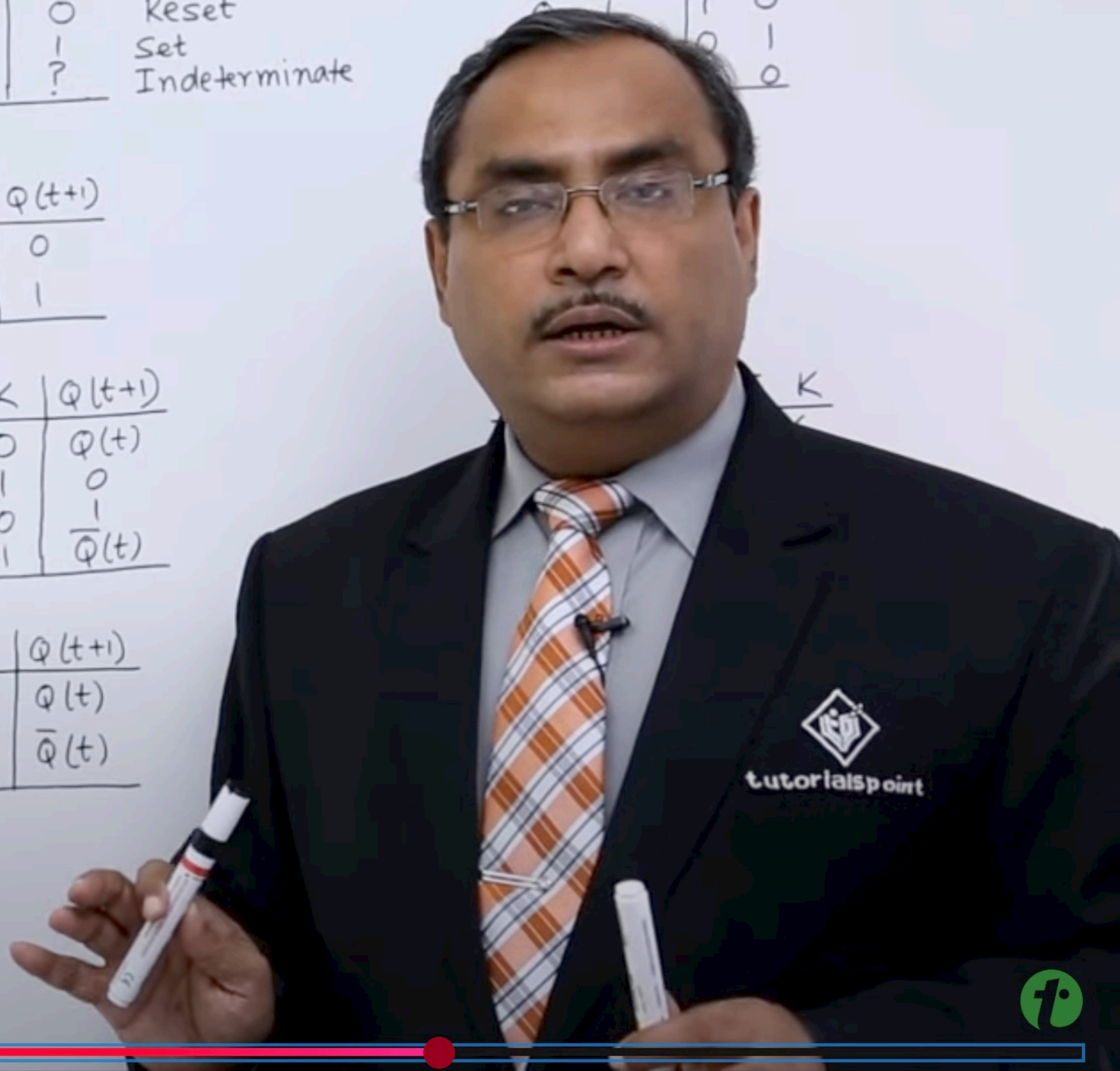
D	Q(t+1)
0	0
1	1

J	K	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	Q-bar(t)

T	Q(t+1)
0	Q(t)
1	Q-bar(t)

Excitation Table

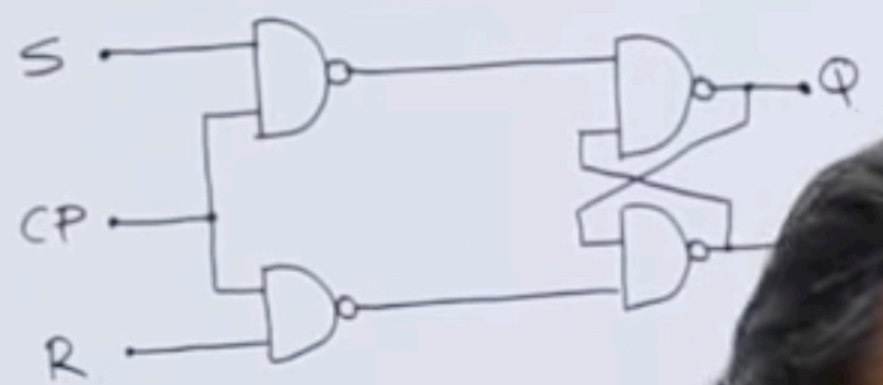
Q(t)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	0	0



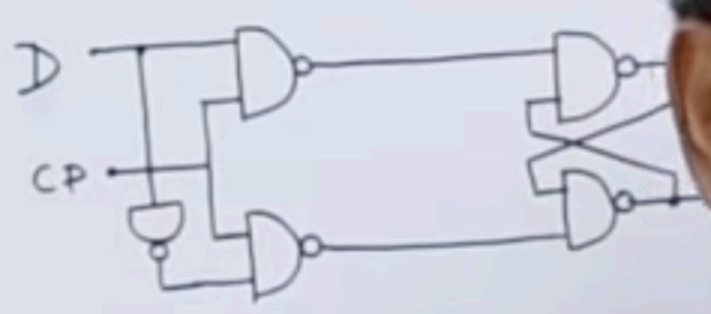
Summary of Flip-Flops

Logic diagram

S-R F/F



D F/F



J-K F/F



T F/F



Characteristic Table

S	R	Q(t+1)
0	0	Q(t) No Change
0	1	0 Reset
1	0	1 Set
1	1	? Indeterminate

Q(t+1)
0
1

K	Q(t+1)
0	Q(t)
1	0

Excitation Table

Q(t)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Q(t)	Q(t+1)	D
0	0	0
0	1	1
1	0	0
1	1	1

Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0